A Novel Spectral Index for Rapid Dust-Proof Net Mapping Based on Sentinel-2 Images

Chaoqun Zhang, Lei Zhou, Mingyi Du, Qiang Chen, and Yang Liu

***Abstract*— Dust-proof net (DPN) is a plastic material used for covering exposed land. While the primary function of DPN is to mitigate dust pollution significantly, it also inadvertently contributes to ecological contaminants. However, DPN mapping has received insufficient attention, constraining comprehensive analysis of DPN distribution and its ecological ramifications. Index-based methods, characterized by their efficiency and intuitive applicability, are more suitable for large-scale DPN mapping than classifier-based approaches. Hitherto, there exists no dedicated spectral index used for DPN mapping, and the spectral characteristics of DPNs remain unclear. In this study, a new spectral index named DPN index (DPNI) is proposed. By analyzing the Sentinel-2-based spectra, the spectral features that distinguish DPNs from other land covers were investigated and used in the DPNI establishment. The results demonstrate that DPNI has significant advantages in suppressing complex backgrounds and enhancing DPNs compared with other indices. The DPNI consistently achieved a mapping accuracy ranging from 93.51% to 98.83% in overall accuracy (OA) and 83.61% to 96.54% in $F1$-score across diverse evaluations. Furthermore, DPNI has superior performance than other indices and similar performance to the random forest (RF) method under a faster premise. The robustness of DPNI was further corroborated across temporal scales, DPN variations, and different urban contexts. DPNI is the first spectral index of the DPN. DPNI is expected to advance automatic and large-scale mapping methods for DPN, thereby informing ecological evaluations on DPN deployments.**

***Index Terms*— Dust-proof net (DPN), DPN index (DPNI), large-scale mapping, Sentinel-2 images, separability analysis.**

I. Introduction

GLOBALLY, urban landscapes are expanding to accommodate rapidly growing urban populations [1]. China, the most populous country in the world, has been undergoing rapid urbanization, accompanied by a large number of urban infrastructure constructions [2]. Such urban development inevitably produces numerous construction sites, subsequently exacerbating dust pollution [3]. A dust-proof net (DPN) is a plastic mesh that has been widely used worldwide as an effective tool for dust reduction. Mandates from both the Ministry of Ecology and Environment (MEE), China, and the Ministry of Housing and Urban-Rural Development (MOHURD), China, emphasize the necessity to deploy DPNs at construction sites. These mandates align with the essence of UN Sustainable Development Goal target 11.6, emphasizing the preservation of urban air quality in the process of urbanization. Although DPN has significantly contributed to air quality improvements, it also has potential environmental impacts. For example, microplastic pollution in the DPN-covered soil can induce alterations in soil attributes (e.g., soil structure and function) [4], [5], thereby potentially compromising food quality [6] and safety [7], [8].

Accurate information on the spatiotemporal distribution of DPNs is essential for urban environmental monitoring and comprehending the ecological ramifications of DPNs. Therefore, there is an exigent need to develop automatic and robust DPN mapping methods to extract spatial and temporal distribution information of DPNs. High-resolution remote sensing image, characterized by its rich spatial details and high timeliness, facilitates the DPN observation [9]. However, the unique structure of the DPN, which is a plastic material and covers other complex materials, makes it challenging to map DPNs using satellite data.

Few studies have focused on DPN mapping. The existing DPN mapping methods can be categorized into visual-interpretation-based and classifier-based approaches. The visual-interpretation-based methods rely on meticulous manual assessments [10], [11]. Although the visual-interpretation-based method can accurately extract the spatial distribution of DPNs, it is labor-intensive and time-consuming. Classifier-based methods have the advantage of automatic mapping, which relies on the feature, classifier, and parameter settings. The spectral differences between DPNs, buildings, and bare soil were initially analyzed and used as features input to the classifier [12]. Furthermore, the object-based features (object-related shape, texture, and color) show superior performance in DPN mapping [13]. These selected features are subsequently processed by the classifier based on specific parameter settings. Zhou et al. [14] used three classifiers [categorical and regression tree (CART), random forest (RF), and support vector machine (SVM)] to map DPN regions and identify optimal parameter settings for each classifier. In contrast to traditional machine learning-based methods, deep learning-based methods have found extensive applications in remote sensing image processing, including change detection [15], [16], [17] and image classification [18], underscoring their value and efficiency. Deep learning-based classifiers exhibit higher

Manuscript received 28 May 2023; revised 11 August 2023 and 3 September 2023; accepted 20 September 2023. Date of publication 27 September 2023; date of current version 10 October 2023. This work was supported by the National Natural Science Foundation (NSFC) through Key Project under Grant 42077439, Grant 42271478, and Grant 41930650. *(Corresponding author: Lei Zhou.)*

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Digital Object Identifier 10.1109/TGRS.2023.3319718

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accuracy over machine learning-based classifiers in extracting DPNs from high-resolution remote sensing images [19], [20]. Although classifier-based methods have demonstrated effectiveness in small-scale or medium-scale mapping, their applicability remains constrained in large-scale, rapid DPN mapping. This limitation is attributed to the fact that the classifier requires exhaustive sample sets and significant computational capacities [21], [22]. Consequently, there is an urgent need to develop a novel method capable of large-scale, rapid DPN mapping to mitigate the shortcomings of classifier-based methods.

Currently, there remains a conspicuous absence of methods for large-scale, rapid DPN mapping. Furthermore, methods developed for rapid and large-scale mapping of other land covers have not been applied to DPN mapping. Index-based methods use combinations of the reflectance of several wavelengths to identify specific land cover. It has demonstrated reliability in large-scale, rapid land cover mapping [23], [24]. It has been successfully used for natural resource monitoring [25], [26] and urban land cover detection [27], [28]. However, there is no spectral index established for the DPN. The potential difficulties are as follows: first, DPN was considered as background in existing studies focusing on spectral indices for urban plastic materials (UPMs) detection [29], [30]. Thus, the spectral consistency and differences between DPNs and other land covers are unclear. In addition, DPN regions on the remote sensing images may be mixed pixels, as porous DPN usually covers different materials (e.g., concrete, cement, and sludge). Thus, a comprehensive understanding of the impact of different materials on the spectral reflectance of DPNs is essential for establishing a robust DPN spectral index, which has not been investigated.

Airborne and satellite hyperspectral data are the most common data used for analyzing the UPMs [29], [31]. For example, the hydrocarbon index (HI), developed using HyMap hyperspectral data, has been successfully used to detect urban hydrocarbon areas [32]. However, it is difficult to obtain airborne hyperspectral data on a global scale. In contrast, high-resolution multispectral data with global observation capabilities are more suitable for large-scale land cover classification [33]. Sentinel-2 data with 10 m spatial resolution has global multispectral and high-resolution observation capabilities, finding extensive applications in urban [34], [35], and natural [36], [37] environment monitoring. In addition, it is common that almost all urban construction sites are covered with DPNs, and the distribution information of DPNs requires continuous updating due to rapid development [12]. With a five-day global revisit cycle, it is expected that Sentinel-2 data can update the spatial distribution information of the DPN timely. In particular, the multispectral information of Sentinel-2 data provides an opportunity to analyze the spectral characteristics of the DPN, which is essential to establish a reliable spectral index [38], [39]. However, a reliable spectral index for identifying DPNs using Sentinel-2 data remains undeveloped.

The main objective of this study is to develop an advanced DPN spectral index and a simple, applicable method for mapping DPNs through the investigation of spectral feature analysis and extraction. First, based on the spectral separability analysis, a novel DPN index (DPNI) is designed to enhance the DPN information and suppress other background land covers (vegetation, cropland, water, artificial surface (AS), blue-roof building (BRB), and UPM). Then, the effectiveness of DPNI is validated by comparing it with other index-based and classifier-based methods. We chose two typical cities in China (Beijing and Zhengzhou) as study areas, as they represent the two main kinds of DPNs [green DPN (GDPN) and blue DPN (BDPN)]. In addition, the selected cities are currently the most representative areas in terms of dense DPN coverage and complex background environments. Therefore, the chosen cities can support the establishment of a generalized and robust DPN spectral index. DPNI will be the first spectral index for DPN mapping. We expect DPNI to advance automatic and large-scale mapping methods for the DPN and support the assessment of the environmental impacts of DPN.

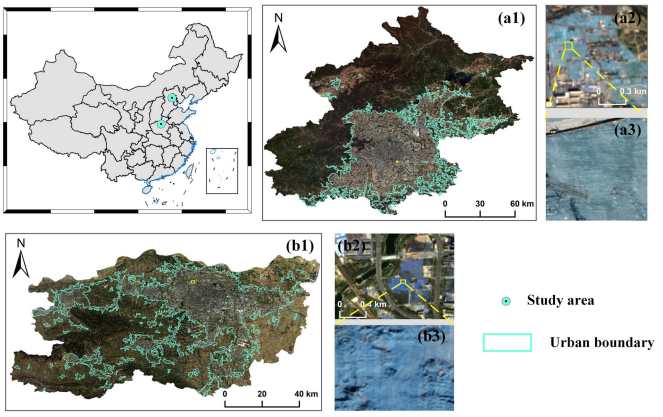


Fig. 1. Location of study areas and DPN examples. (a1) and (b1) Sentinel-2 images covering study areas. (a2) and (b2) DPN examples in Sentinel-2 in Beijing and Zhengzhou. (a3) and (b3) DPN examples in Google Earth images in Beijing and Zhengzhou. The urban boundary data are obtained from the GUB product [40].

II. Study Areas and Data

A. Study Areas

In China, DPNs are chiefly classified as GDPN and BDPN. Their texture characteristics are similar, but they differ in color. Beijing and Zhengzhou were chosen as study areas because they contain both GDPNs and BDPNs. Fig. 1 illustrates the locations of Beijing and Zhengzhou. Beijing is a megalopolis located in the northeast of China. Typically, the DPNs in Beijing are GDPNs, as shown in Fig. 1(a3). Zhengzhou is located in Henan, China. The DPNs in Zhengzhou are usually BDPNs, as shown in Fig. 1(b3). Both cities exhibit a dense coverage of DPNs and have complex background environments, complicating DPN mapping. Therefore, a spectral index effective for these study areas is expected to have a robust capability for large-scale DPN detection.

B. Data

1) Sentinel-2 Data: As shown in Table I, Sentinel-2A is equipped with advanced multispectral sensors that provide four 10 m spatial resolution bands (bands 2, 3, 4, and 8),

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TABLE I
WAVELENGTH AND SPATIAL RESOLUTION OF SENTINEL-2A DATA

Sentinel-2A bands	Central wavelength (nm)	Pixel Size (m)	Description
Band 1	443.9	60	Coastal aerosol
Band 2	496.6	10	Blue
Band 3	560	10	Green
Band 4	664.5	10	Red
Band 5	703.9	20	Red Edge 1
Band 6	740.2	20	Red Edge 2
Band 7	782.5	20	Red Edge 3
Band 8	835.1	10	NIR
Band 8A	864.8	20	Red Edge 4
Band 9	945	60	Water vapor
Band 11	1613.7	20	SWIR 1
Band 12	2202.4	20	SWIR 2

six 20 m spatial resolution bands (bands 5, 6, 7, 8A, 11, and 12), and three 60 m spatial resolution bands (bands 1, 9, and 10). In this study, Sentinel-2A Multispectral Instrument (MSI) data in Google Earth Engine were used, which provides high-resolution satellite multispectral data. Level-2A products were used in this study. In Level-2A products, the radiometric measurements for each pixel are provided as surface reflectance. Note that band 10 in Sentinel-2A (the band used for atmospheric correction) is unavailable in Level-2A products as they have been atmospherically corrected.

Remote sensing images in Beijing and Zhengzhou were collected from each of the four seasons in 2020. We used median values of each pixel with less than 10% cloud coverage to avoid the influence of the cloud. The obtained cloud-free images were able to satisfy the demands of this study. Furthermore, the band images with a spatial resolution of fewer than 10 m were resampled to 10 m to ensure the consistency of spatial resolution.

2) Field Spectral Data: To validate the effectiveness of the proposed index for the DPN covering different construction materials, field investigations were conducted in Beijing and Zhengzhou on April 16th, 2021, and December 2nd, 2021, respectively. Spectral measurements were performed for six types of DPNs under clear-sky conditions. The analytical spectral device (ASD) spectroradiometer was used to obtain field spectral data, which has a wavelength range from 350 to 2500 nm. Spectral ranges (from 1350 to 1450 nm, from 1800 to 1999 nm, and from 2401 to 2500 nm) were removed due to noise caused by atmospheric water absorption. The field spectral reflectance is produced using white reference spectral panels. The resampled Sentinel-2 spectral reflectance was obtained using the spectral response function of the Sentinel-2 sensor provided by ENVI5.3 [41].

III. METHOD

Fig. 2 illustrates the workflow of DPN mapping. First, the spectral characteristics of the DPNs were investigated in detail using Sentinel-2 spectral reflectance. Then, an effective DPNI was established based on the spectral separability analysis between DPN and other land covers. Finally, the reliability of DPN was evaluated by comparing it with other index-based and classifier-based methods.

A. Separability Analysis of DPN Spectra

The Sentinel-2 spectral reflectance of eight land covers was analyzed to develop an effective DPN spectral index. The eight land covers include two DPN types (GDPN and BDPN), five land covers (cropland, vegetation, water, UPM, and AS), and a special category (BRB). The BRB typically represents factory buildings and is prevalent in urban areas. It has been demonstrated that the spectral characteristics of BRB are similar to plastic materials and different from AS [28], [42]. Therefore, it was specially considered for further analysis. Note that it is challenging to sample UPM through visual interpretation of remote sensing images. Here, we utilized UPM sampling locations confirmed by Guo and Li [29]. Most sampled UPMs were synthetic turf, plastic shed covers, and plastic tile roofs. Enough samples can be used to analyze the separability of the DPN spectra. With the help of Google Earth images, 1000 DPN samples and 1000 samples for each non-DPN category (except UPMs) were selected under four seasons in each study area. In total, 2000 DPN samples (1200 GDPN samples and 800 BDPN samples), 10 000 non-DPN samples, and 30 UPM samples were collected. Fig. 3 shows the average spectral reflectance across the eight land covers.

To select the appropriate band combination, the separability between each two land covers at a specific band or band combinations was measured using the M-index. M-index focuses on the relative position and distribution of histograms between two land covers. The degree of separation depends on the variance between and within classes. The higher value of the M-index indicates better separability. Moreover, $M > 1.0$ means excellent separability, while $M < 1.0$ means bad separability. The M-index is calculated as follows:

$$M = \frac{|\mu 1 - \mu 2|}{\sigma 1 + \sigma 2} \tag{1}$$

where $\mu 1$ and $\mu 2$ are average values of two indices, and $\sigma 1$ and $\sigma 2$ are standard deviations of two indices.

The results of the M-index showed that it is challenging to distinguish DPNs from BRBs, AS, and UPMs by a single band (see Table II). As shown in Fig. 3, $B7–B2$ is a good indicator for highlighting DPNs and suppressing BRBs. This is because the reflectance of $B7$ of BRB is lower than that of $B2$, while the other land covers (except water) are opposite. Table II illustrates that $B7–B2$ effectively separates BRBs from GDPNs and BDPNs, with respective M-indexes of 1.40 and 2.41. DPNs show a higher spectral reflectance in $B2$ and stronger absorption in $B4$ than AS (see Fig. 3), making $B2–B4$ essential for distinguishing DPNs from AS. As shown in Table II, the M-index of $B2–B4$ exceeds 2.02, indicating that $B2–B4$ is an effective indicator for distinguishing AS from both GDPNs and BDPNs. It seems more challenging to distinguish DPNs from UPMs, as they have similar spectral reflectance in the spectral range from near infrared (NIR) to short-wave infrared (SWIR). In the visible spectral range, the reflectance of DPNs in $B2$ and the absorption of DPNs in $B4$ are higher than those of UPMs. Therefore, with an M-index

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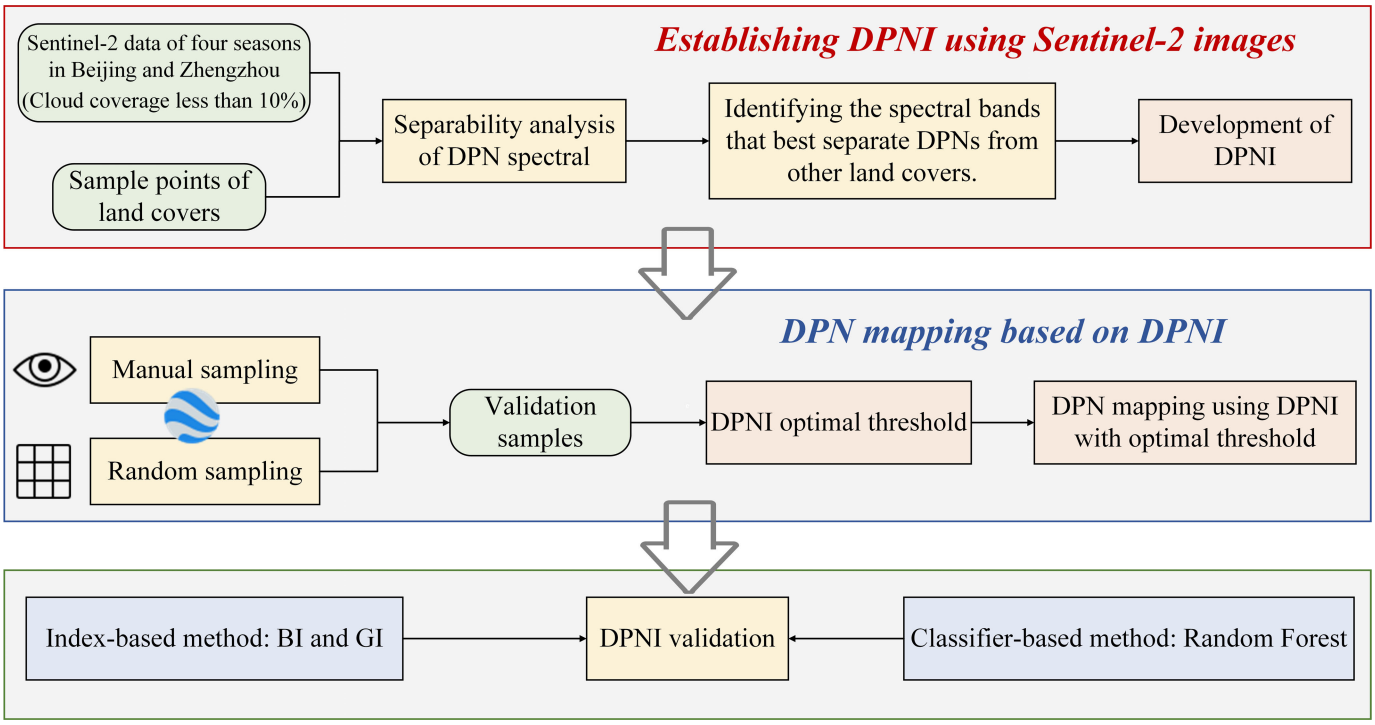


Fig. 2. Flowchart of the DPNI establishment.

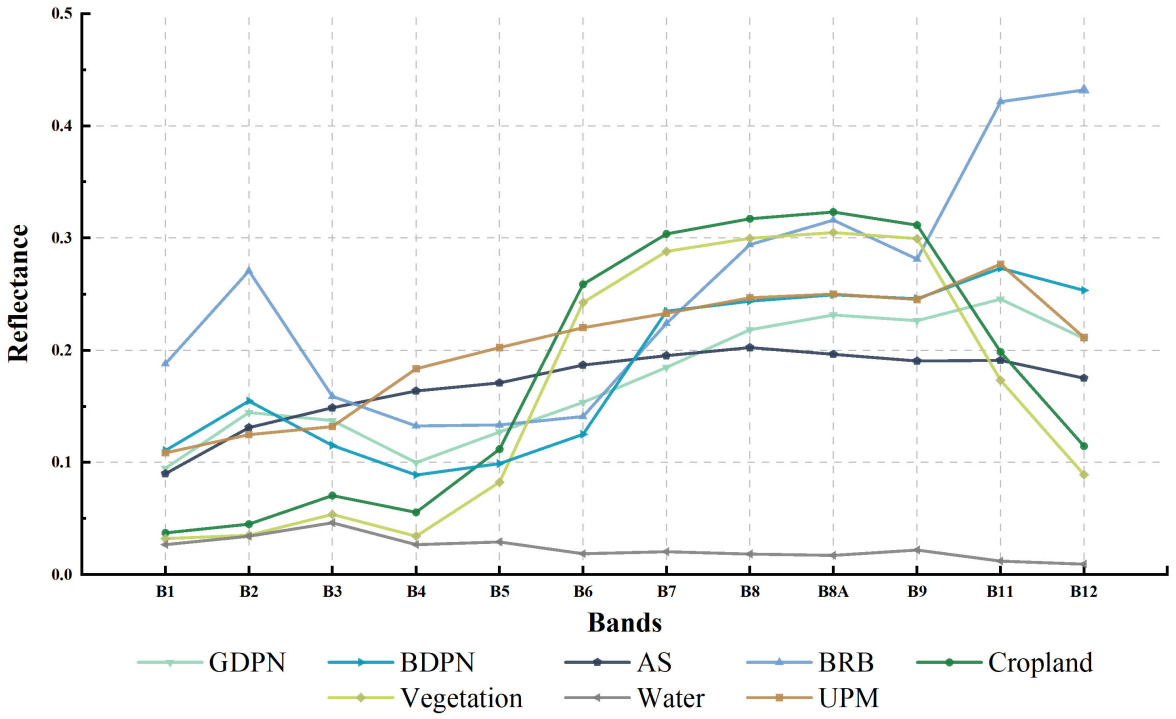


Fig. 3. Sentinel-2 average reflectance spectra of various land cover types.

exceeding 1.41, $B2$–$B4$ is also effective in distinguishing UPMs from both GDPNs and BDPNs.

However, the $B7$–$B2$ and $B2$–$B4$ indicators are insufficient to suppress vegetation, cropland, and water. By analyzing the spectral reflectance in Fig. 3, it can be found that DPNs have a certain degree of spectral differentiation with vegetation, cropland, and water. $B2$ and $B12$ effectively distinguish DPNs from vegetation, cropland, and water, as the M-index of $B2$ is greater than 3.73, while that of $B12$ is greater than 1.38.

B. Development of the DPN Index

Based on the above separability analysis, $B7$–$B2$ is an effective indicator to distinguish DPNs from BRBs. $B2$–$B4$ is

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TABLE II
SEPARABILITY MEASURES BETWEEN DPN (GDPN AND BDPN) AND OTHER LAND COVERS WITH *M*-INDEX IN EACH BAND OR BAND COMBINATION

	B1	B2	B3	B4	B5	B6	B7	B8	B8A	B9	B11	B12	B2-B4	B7-B2
GDPN&AS	0.11	0.21	0.17	0.86	0.59	0.44	0.14	0.17	0.43	0.55	0.52	0.34	2.02	0.38
GDPN&BRB	1.41	2.18	0.38	0.62	0.11	0.21	0.54	1.03	1.12	0.72	1.36	1.50	2.04	1.40
GDPN&Cropland	2.37	3.73	2.24	1.29	0.35	1.21	1.18	0.95	0.90	1.02	0.61	1.38	2.21	2.21
GDPN&Vegetation	2.65	5.13	3.49	2.90	1.32	1.15	1.06	0.81	0.73	0.79	0.83	1.86	2.42	2.24
GDPN&Water	2.84	4.15	2.78	2.79	3.10	4.31	4.58	5.38	5.72	4.23	4.42	4.15	2.05	1.46
GDPN&UPM	0.13	0.21	0.12	0.96	0.76	0.61	0.42	0.21	0.10	0.11	0.18	0.07	1.41	0.59
BDPN&AS	0.47	0.36	0.54	1.11	1.06	0.92	0.53	0.49	0.69	0.91	0.99	0.89	2.41	0.30
BDPN&BRB	1.13	1.94	0.84	0.95	0.69	0.31	0.16	0.75	0.94	0.49	1.36	1.35	1.49	2.41
BDPN&Cropland	2.78	3.86	1.87	1.19	0.36	1.69	0.72	0.75	0.77	0.83	1.31	2.57	2.77	2.00
BDPN&Vegetation	3.04	5.18	3.41	3.39	0.60	1.69	0.58	0.59	0.58	0.61	1.50	3.32	3.14	2.03
BDPN&Water	3.21	4.25	2.57	3.16	2.82	4.57	7.08	7.35	7.20	5.09	8.02	7.45	2.81	3.42
BDPN&UPM	0.03	0.31	0.21	1.19	1.15	0.99	0.10	0.06	0.10	0.10	0.08	0.62	1.65	0.23

the best indicator to distinguish DPNs from AS and UPMs. $B2$ and $B12$ are the best spectral bands to distinguish DPNs from vegetation, cropland, and water. Therefore, we designed an advanced DPNI to highlight DPNs, as shown in the following equation:

$$\mathrm{DPNI} = 100 \times B2 \times B12 \times \frac{B7 - B2}{B7 + B2} \times \frac{B2 - B4}{B2 + B4} \quad (2)$$

where the multiplier 100 stretches the DPNI to a more extensive range linearly. The band combinations of $B2$–$B4$ and $B7$–$B2$ are expressed as normalized forms. In the equation, the two normalized band combinations are multiplied by each other, which can separate DPNs from BRBs, AS, and UPMs. $B2$ and $B12$ are the best spectral bands that can separate DPNs from vegetation and cropland, and they are used as multipliers to suppress vegetation and cropland. Moreover, $B2$ and $B12$ are also effective for separating DPNs from water. The purpose of all these designs is to highlight DPNs and suppress backgrounds, making the constructed DPNI an effective DPN spectral index.

C. DPN Mapping Method Using Proposed DPNI

A simple threshold segmentation method was used to identify DPN pixels. A pixel is recognized as a DPN pixel if its DPNI value exceeds a given threshold. Otherwise, the pixel is considered as background.

It is essential to determine an optimal threshold for DPN mapping using DPNI. In this study, a quantitative evaluation method based on random sampling was used to determine the optimal threshold for DPN mapping. The process of this method is shown in Fig. 4. The first step is sample selection. We randomly selected 1200 DPN samples from the 2000 DPN samples mentioned in Section III-A. To ensure randomness, a fishnet-based random sampling method was employed for non-DPN samples. First, a fishnet of size 25 × 25 was established in the study areas, with a total of 625 fishnet units. Three sample points were randomly generated in each fishnet unit, and each sample point corresponds to a pixel of 10 m resolution. It should be noted that they may be either DPNs or non-DPNs. Therefore, it is necessary to check the accuracy of the samples manually. Using Google Earth images,

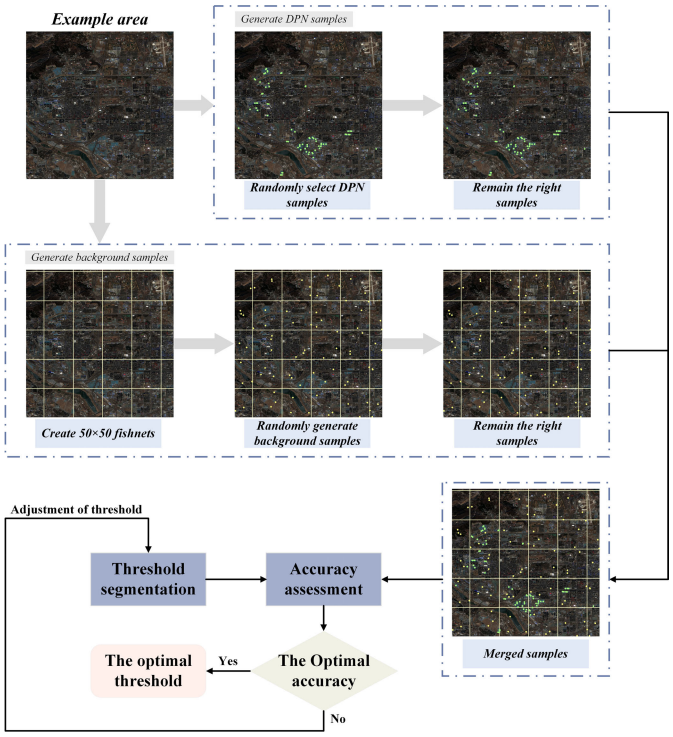


Fig. 4. Flowchart of determining the optimal threshold.

we determined the land cover types of sample points and removed the wrong samples. Then, all correct samples were merged. Finally, the threshold is initiated from the minimum value of DPNI and is incremented by 0.01 at each step until the maximum value of DPNI, thereby obtaining the segmentation results of all the sample points under different thresholds. The optimal threshold was determined by the accuracy of the segmentation results.

D. Evaluation Method

1) Visual Evaluation: Visual evaluation can intuitively assess the performance of the index in highlighting the DPN and suppressing the background. All index results were displayed using histogram percentage stretching to ensure a fair

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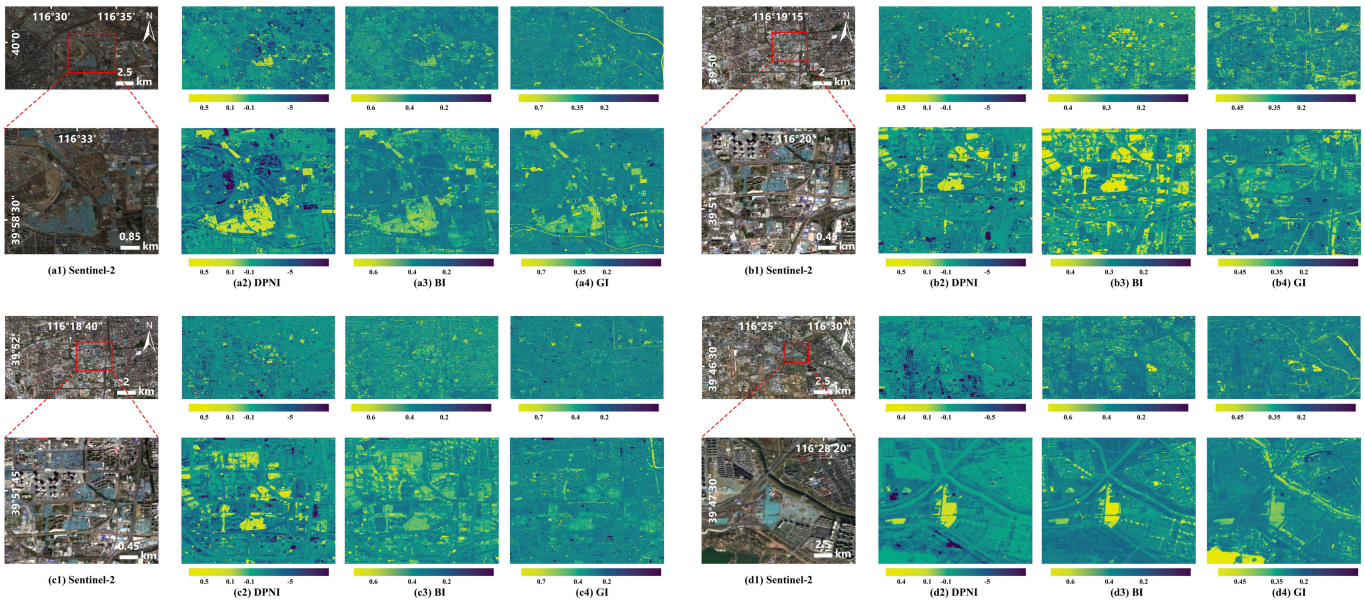


Fig. 5. Visual evaluation results of the three indices in Beijing. (a1)–(d1) Sentinel-2 images in spring, summer, autumn, and winter, respectively. (a2)–(d2) DPNI. (a3)–(d3) BI. (a4)–(d4) GI.

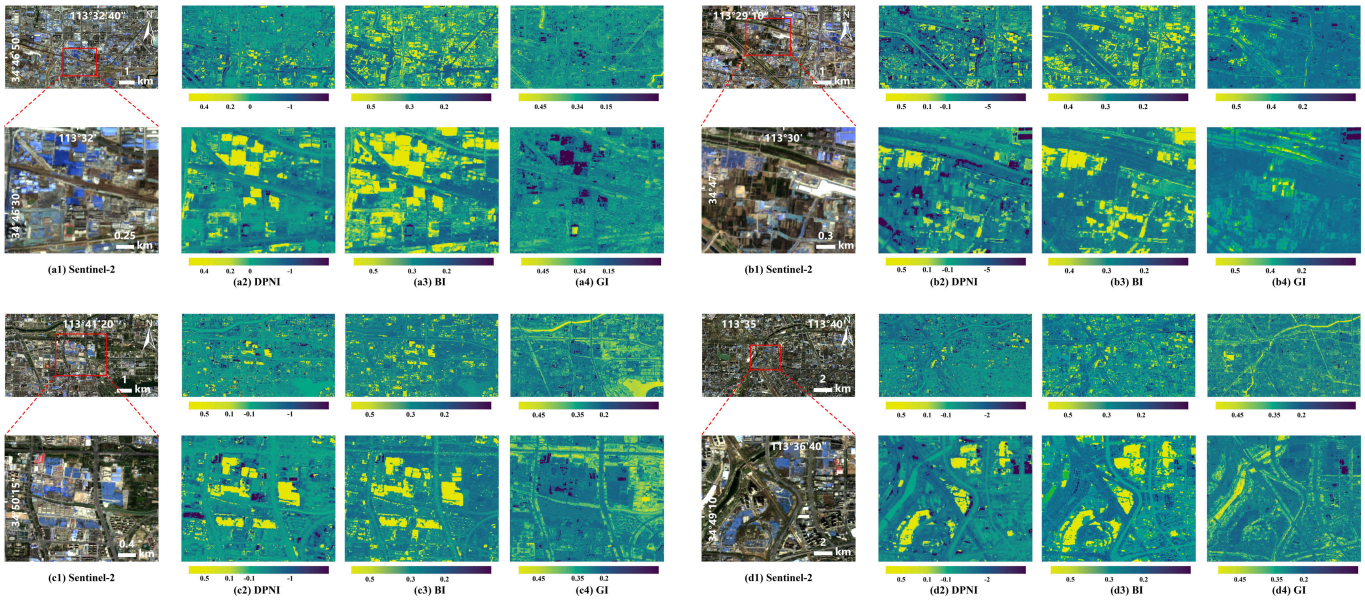


Fig. 6. Visual evaluation results of the three indices in Zhengzhou. (a1)–(d1) Sentinel-2 images in spring, summer, autumn, and winter, respectively. (a2)–(d2) DPNI. (a3)–(d3) BI. (a4)–(d4) GI.

comparison and avoid biases due to unavoidable noise. The performance of each index was comprehensively evaluated through visual evaluation in two kinds of spatial scales (large-scale and detailed areas).

2) Comparison With Existing Indices: No existing indices are purely suitable for DPN monitoring. We selected two similar color-based indices, the green index (GI) and the blue index (BI), to compare with DPNI, as GDPNs and BDPNs are two main types of DPNs. These two indices were initially developed based on Landsat TM data for soil classification. The GI and BI are calculated as follows:

$$BI = \frac{B2}{B2 + B3 + B4} \tag{3}$$

$$GI = \frac{B3}{B2 + B3 + B4}. \tag{4}$$

The threshold segmentation method for GI and BI is the same as DPNI because their high values are related to DPN.

3) Comparison With Existing DPN Mapping Method: RF has been demonstrated to be effective for DPN mapping [12], [14]. The DPN mapping method based on DPNI is compared with RF to evaluate the advantages of DPNI. In this study, the parameters of the RF classifier were set to 50 trees and four random seeds, as recommended by Zhou et al. [14]. $B2$, $B6$, $B12$, NDVI, and NDWI on Sentinel-2 data were selected as input features for the RF classifier. In this regard, $B2$, $B6$, and $B12$ have been substantiated as the spectral bands with

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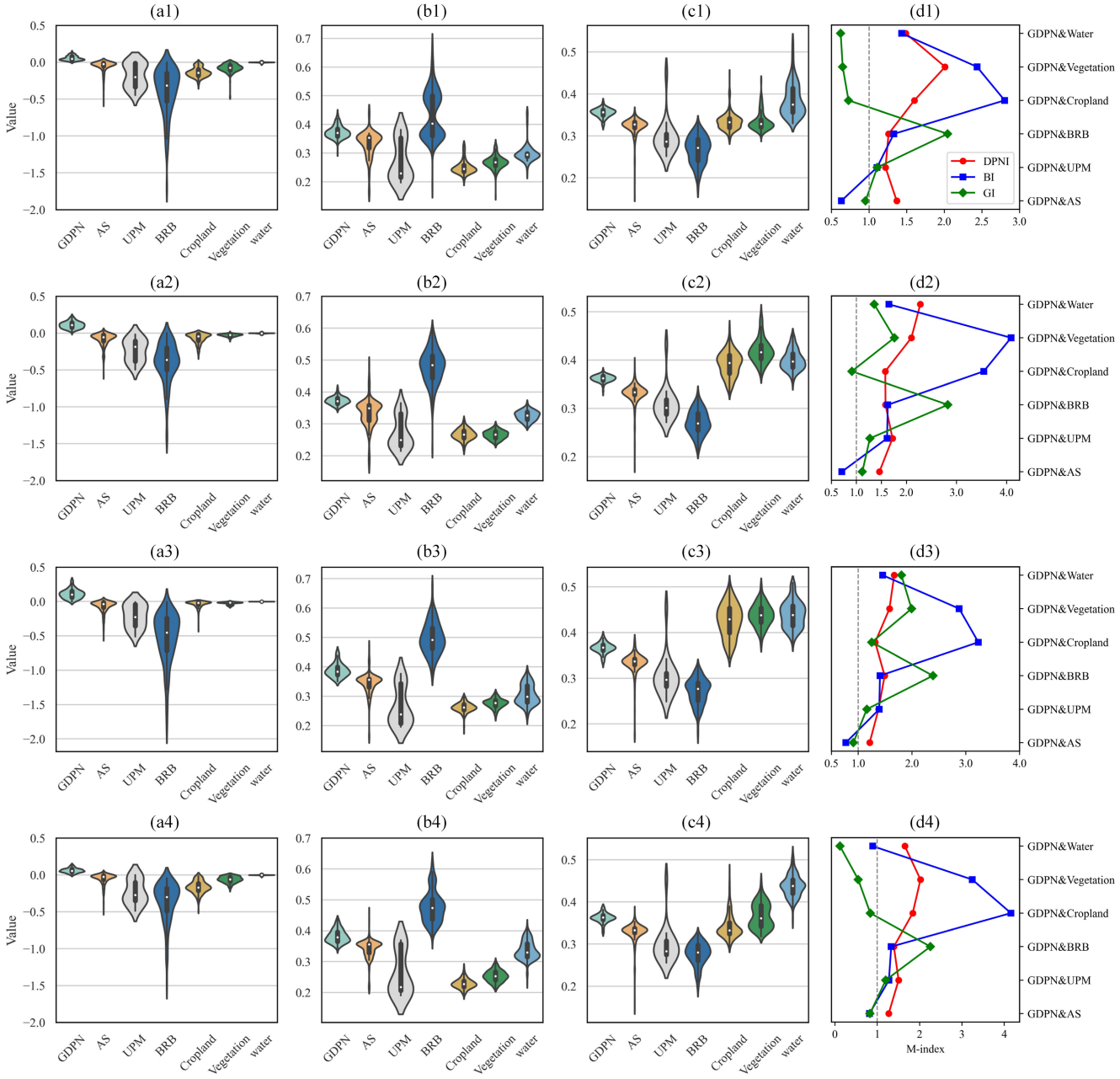


Fig. 7. Results of statistical distribution and separability measures in Beijing. (a1)–(a4) Statistical distribution of DPNI for different land cover types in spring, summer, autumn, and winter, respectively. (b1)–(b4) Statistical distribution of BI for different land cover types in spring, summer, autumn, and winter, respectively. (c1)–(c4) Statistical distribution of GI for different land cover types in spring, summer, autumn, and winter, respectively. (d1)–(d4) M-index of three indices between GDPN and other land covers in spring, summer, autumn, and winter, respectively.

substantial differences between DPNs and other types of land covers [12]. In addition, due to the abundant presence of vegetation and water bodies within cities, the normalized difference vegetation index (NDVI) and the normalized difference water index (NDWI) also serve as important features in distinguishing DPNs from other land covers. We randomly selected 80% of the merged samples mentioned in Section III-C as the training set to train the RF model, while 20% were designated as the validation set to quantitatively assess the performance of the trained RF model.

4) Assessment Indices: Overall accuracy (OA), producer accuracy (PA), user accuracy (UA), and $F1$-score were selected to measure mapping accuracy. These metrics are calculated as follows:

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FN} + \text{FP} + \text{TN}} \tag{5}$$

$$\text{UA} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{6}$$

$$\text{PA} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{7}$$

$$\text{F1-Score} = \frac{2 \times UA \times PA}{UA + PA} \tag{8}$$

where TP is the number of pixels in the category DPN that are present in both ground truths and predictions. TN denotes the number of pixels in the category DPN that are only present in

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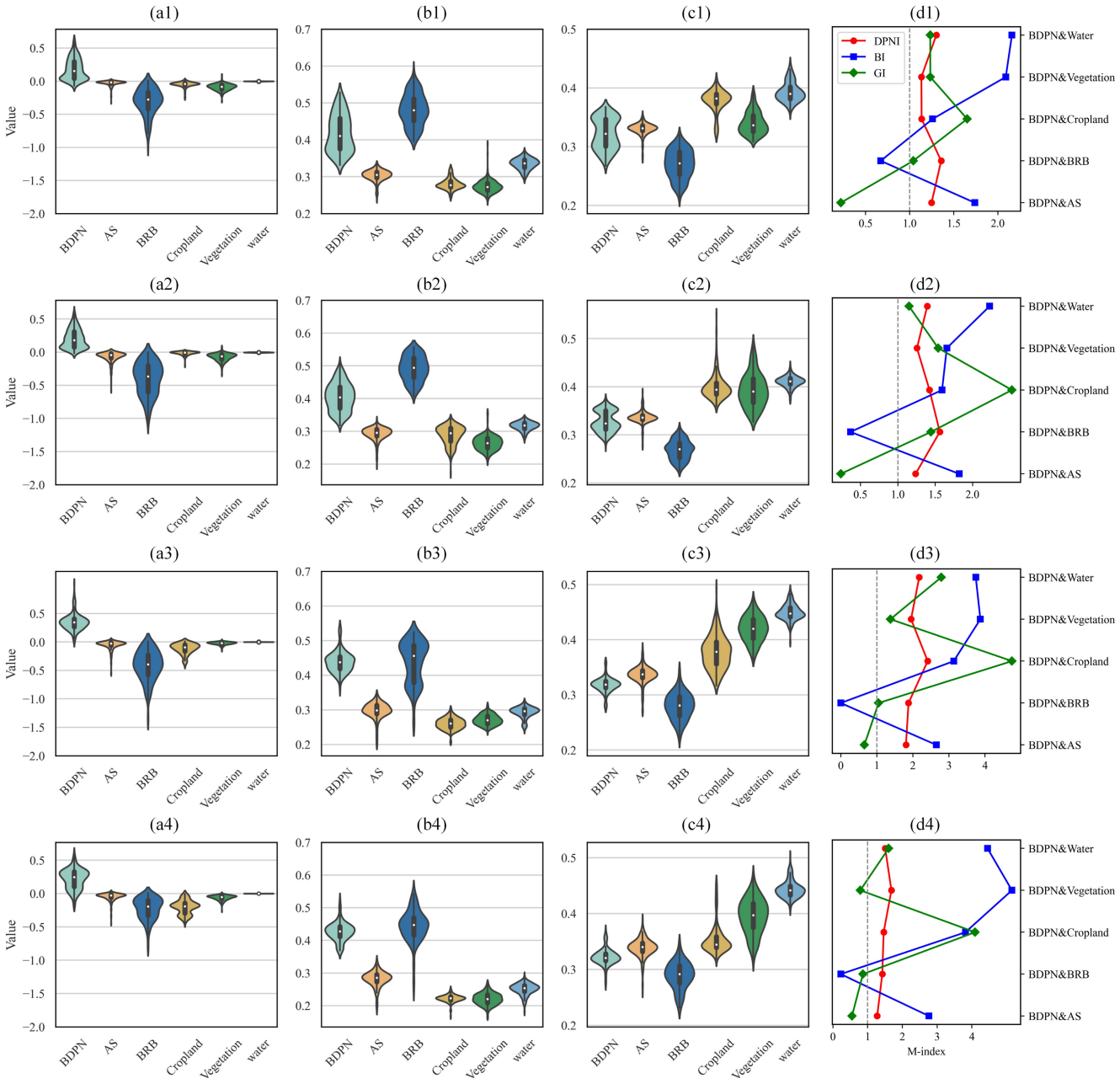


Fig. 8. Results of statistical distribution and separability measures in Zhengzhou. (a1)–(a4) Statistical distribution of DPNI for different land cover types in spring, summer, autumn, and winter, respectively. (b1)–(b4) Statistical distribution of BI for different land cover types in spring, summer, autumn, and winter, respectively. (c1)–(c4) Statistical distribution of GI for different land cover types in spring, summer, autumn, and winter, respectively. (d1)–(d4) M-index of three indices between BDPN and other land covers in spring, summer, autumn, and winter, respectively.

ground truths but not in predictions. FP indicates the number of pixels in the category DPN that are only present in predictions but not in ground truths. FN refers to the number of pixels in the category DPN that are not present in both ground truths and predictions.

IV. Results

A. Visual Evaluation Results

Fig. 5(a1)–(d1) shows the Sentinel-2 images of Beijing in spring, summer, autumn, and winter, respectively. The visual evaluation results for BI and GI are shown in Fig. 5(a3)–(d3) and (a4)–(d4), respectively. Most DPN pixels exhibited high BI values, which were highlighted in the visual evaluation results. However, many background pixels were confused with the DPN pixels in the visual evaluation results of BI. For BI, many AS pixels, especially the BRB, were wrongly highlighted [see Fig. 5(b3)]. This is because the blue band reflectance of the BRB is higher than the red and green band reflectance. The vegetation and water pixels were highlighted by GI, rather than DPN pixels. While some DPN pixels are highlighted in the spring, GI lacks robustness for DPN pixels across all seasons. This is because the green band reflectance of the vegetation and water is higher than the red and blue

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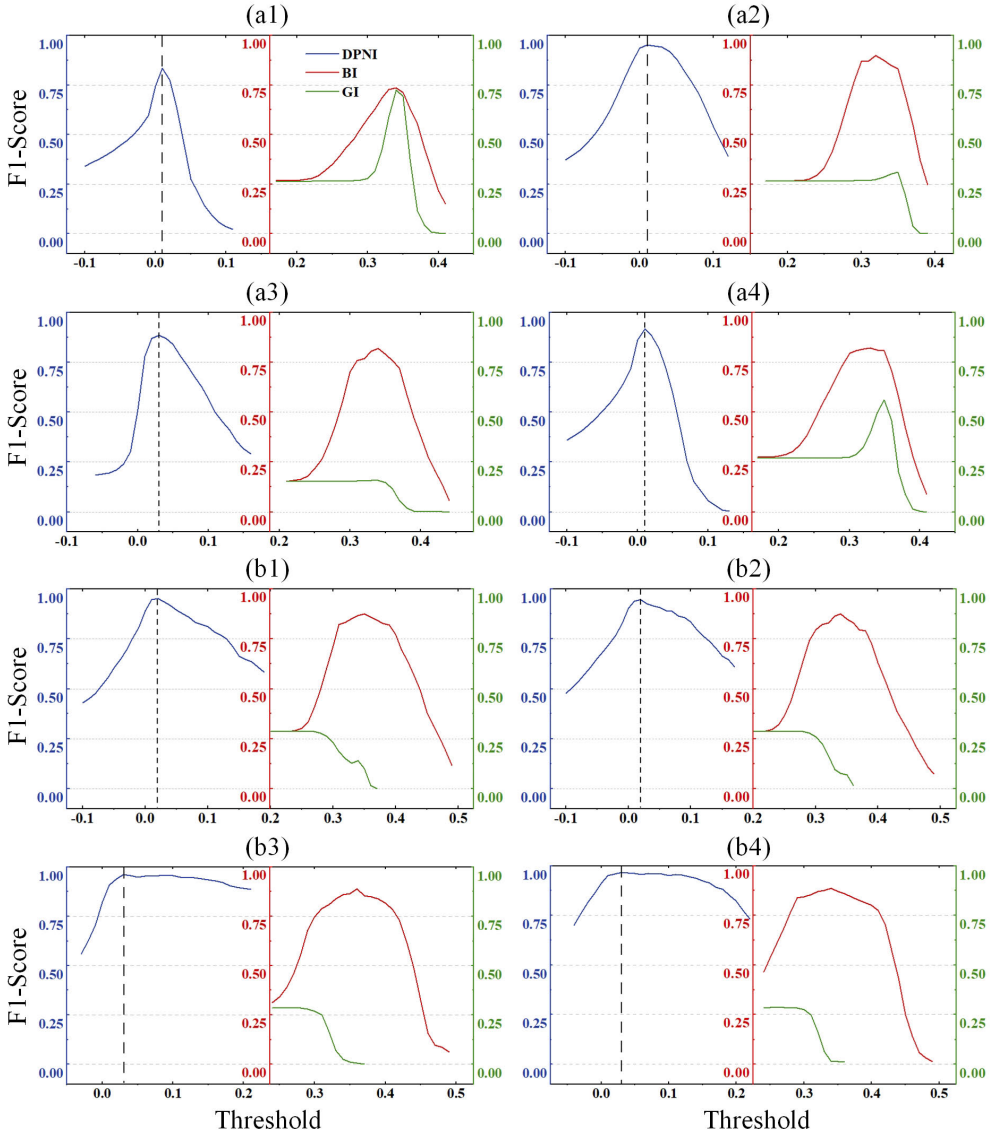


Fig. 9. Accuracy assessment curves of the three indices at different seasons and thresholds in (a1)–(a4) Beijing and (b1)–(b4) Zhengzhou. (a1)–(a4) Spring, summer, autumn, and winter in Beijing, respectively. (b1)–(b4) spring, summer, autumn, and winter in Zhengzhou, respectively.

band reflectance (see Fig. 3). DPNI effectively suppressed most background pixels and highlighted the DPN pixels in all seasons.

Fig. 6(a1)–(d1) shows the Sentinel-2 images of Zhengzhou in spring, summer, autumn, and winter, respectively. BI achieved a similar performance as that in Beijing. The AS pixels, especially the BRB pixels, were still not suppressed, as shown in Fig. 6(a3). GI seemed ineffective in highlighting the BDPNs. The vegetation and water pixels were wrongly highlighted in GI. DPNI consistently and accurately highlighted the DPN pixels in Zhengzhou across all seasons, maintaining its effectiveness among the three indices.

B. Separability Analysis

Fig. 7 shows the statistical distribution of all samples in Beijing for the three indices. Fig. 7(d1)–(d4) shows the separability (M-index) between GDPNs and five land cover types in Beijing. The results showed that DPNI is the best index that can distinguish GDPNs from other land cover types. GDPNs have higher DPNI values than other land covers in four seasons. Across all seasons, DPNI consistently exhibits an M-index greater than 1, illustrating a robust separability between DPNs and other land covers. Compared to the DPNI, none of the other indices have satisfactory separability.

BI could separate GDPNs from vegetation and cropland. However, the BI values of BRB exceed those of the GDPNs in all seasons, indicating that BI does not effectively suppress BRB. Moreover, with an M-index less than 1, BI does not effectively separate GDPNs from AS. Given the similar GI value distribution, GI shows little capacity for distinguishing GDPNs from other land covers. During the summer and autumn, GI emphasizes vegetation and cropland more than GDPNs. Fig. 8 shows the statistical distribution of all samples in Zhengzhou for the three indices. BI is effective in separating BDPNs from the other four land covers. However, BI fails to

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TABLE III
MAPPING ACCURACY USING RF, DPNI, BI, AND GI AT OPTIMAL THRESHOLDS IN SPRING, SUMMER, AUTUMN, AND WINTER, RESPECTIVELY

Study area	Method	UA (%)	PA (%)	F1-Score (%)	OA (%)	Threshold
Spring Beijing	DPNI	92.76	86.10	89.30	95.46	0.01
	BI	64.18	86.45	73.67	90.59	0.34
	GI	64.50	82.95	72.57	90.45	0.34
	RF	**96.92**	**96.02**	**96.47**	**97.53**	
Summer Beijing	DPNI	95.86	94.41	95.13	98.52	0.01
	BI	83.46	97.58	89.97	96.67	0.32
	GI	19.39	78.25	31.07	46.82	0.35
	RF	**96.65**	**97.24**	**96.94**	**98.68**	
Autumn Beijing	DPNI	93.00	84.29	88.43	98.17	0.03
	BI	70.77	97.28	81.93	96.44	0.34
	GI	8.62	96.07	15.82	15.14	0.34
	RF	**95.78**	**94.88**	**95.32**	**98.72**	
Winter Beijing	DPNI	96.56	87.52	91.82	97.58	0.01
	BI	71.63	96.43	82.20	93.51	0.33
	GI	43.22	79.64	56.04	80.58	0.35
	RF	**98.52**	**97.81**	**98.16**	**98.58**	
Spring Zhengzhou	DPNI	97.22	93.33	95.24	98.44	0.02
	BI	78.88	98.33	87.54	95.33	0.35
	GI	16.70	100	28.62	16.84	0.27
	RF	**98.34**	**96.67**	**97.50**	**99.13**	
Summer Zhengzhou	DPNI	94.39	95.02	94.70	98.22	0.02
	BI	82.40	93.35	87.54	95.56	0.34
	GI	16.80	100	28.76	17.21	0.28
	RF	**97.33**	**97.33**	**97.33**	**98.78**	
Autumn Zhengzhou	DPNI	93.97	98.34	96.10	98.67	0.03
	BI	82.43	96.68	88.99	96.00	0.36
	GI	16.77	100	28.72	17.00	0.26
	RF	**97.67**	**97.67**	**97.67**	**98.89**	
Winter Zhengzhou	DPNI	95.13	97.99	96.54	98.83	0.03
	BI	79.84	99.33	88.52	95.72	0.34
	GI	16.66	100	28.56	16.80	0.27
	RF	**98.33**	**100**	**99.16**	**99.89**	

separate BDPNs from BRB, as the M-index values are less than 1 in all four seasons. GI can highlight vegetation and cropland rather than BDPNs and is ineffective in distinguishing BDPNs from AS and BRB. DPNI can highlight BDPNs and successfully separate BDPNs from other land cover types in all seasons.

More importantly, DPNI is effective for two kinds of DPNs, an efficacy unmatched by other indices.

C. DPN Mapping Results

Three indices and RF were used to map the DPN distribution of the study areas. The quantitative assessment results of the three indices at different seasons and thresholds are shown in Fig. 9. Fig. 9(a1)–(a4) show the quantitative assessment results for spring, summer, autumn, and winter in Beijing, respectively. Specifically, BI performed well in all four seasons. GI was effective in the spring and winter but performed poorly in the summer and fall. This is because most vegetation in summer and autumn is green, resulting in the confusion of DPNs with vegetation in the GI mapping results. Conversely, DPNI achieved the highest mapping accuracy across all seasons, highlighting its optimal performance in DPN mapping.

Fig. 9(b1)–(b4) shows the quantitative assessment results for spring, summer, autumn, and winter in Zhengzhou, respectively. Similar to Beijing, DPNI achieved better mapping results than BI in all four seasons. In Zhengzhou, the inefficiency of the GI was apparent throughout the seasons, primarily due to the predominantly blue color of DPNs. Overall, DPNI has the advantage of being effective in different seasons and areas. DPNI could achieve high mapping accuracy even in complex urban environments.

Table III illustrates the DPN mapping accuracy using each index at the optimal threshold and the DPN mapping accuracy using RF. It can be seen that the optimal threshold of DPNI for DPN mapping has stable values over different areas and times. In Beijing, the optimal threshold is 0.01 for spring, summer, and winter and 0.03 for autumn. In Zhengzhou, the optimal threshold is 0.02 in spring and summer, while 0.03 in autumn and winter. DPNI achieved satisfactory mapping accuracy in all cases, with an OA of 93.51%–98.83% and an $F1$-score of 83.61%–96.54%. DPNI had a better performance compared to other indices. In some cases, the mapping accuracy of DPNI

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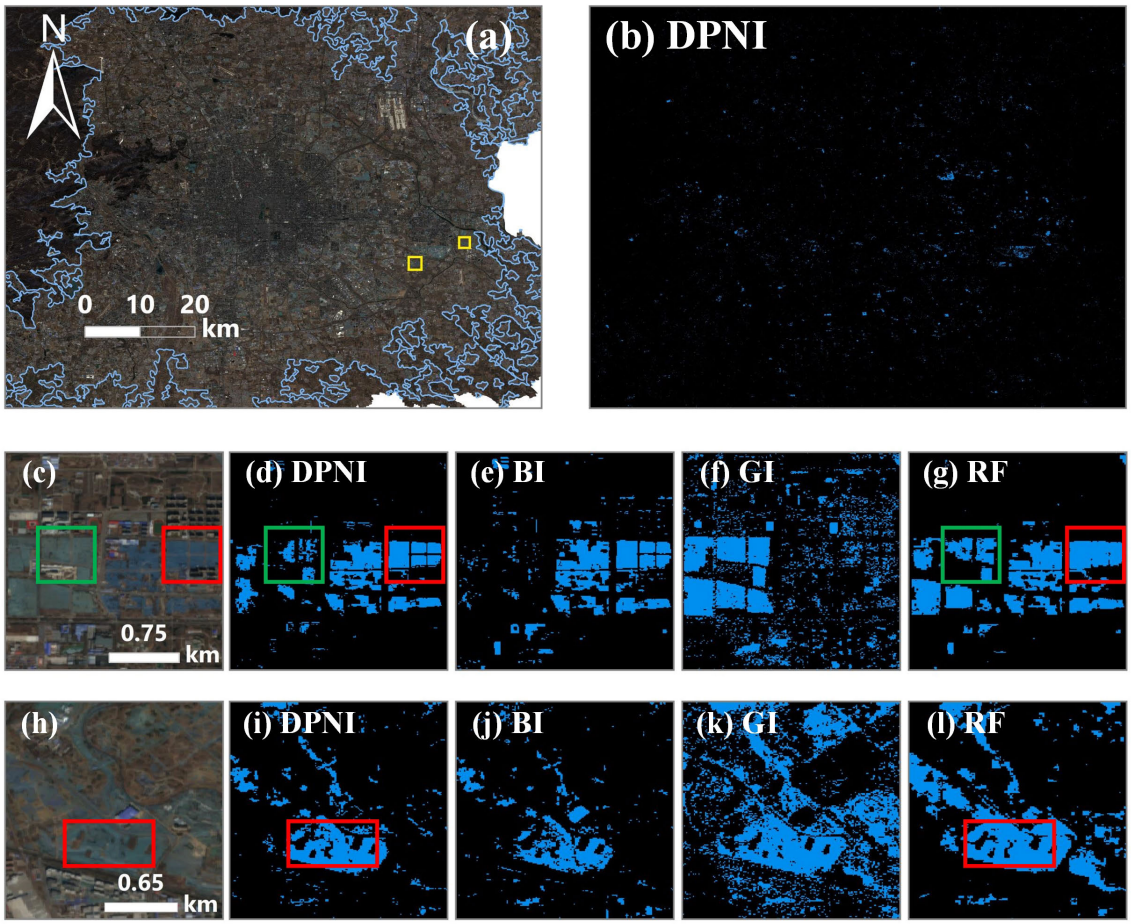


Fig. 10. DPN mapping results using different indices and the RF method in Beijing. The yellow rectangle in (a) is the example region. (c) and (h) Sentinel-2 images of example regions. (b), (d), and (i) DPNI. (e) and (j) BI. (f) and (k) GI. (g) and (l) RF.

is close to that of RF. Generally, DPNI has advantages in DPN mapping.

Figs. 10 and 11 show the DPN mapping results using RF and all indices at the optimal threshold in Beijing and Zhengzhou, respectively. Overall, DPNI could accurately map most DPNs, as shown in Figs. 10(b) and 11(b), and was more effective than BI and GI. For example, BI was ineffective in mapping areas sparsely covered with DPNs [see Fig. 10(e)], and BI confused BRBs and DPNs [see Figs. 10(j) and 11(e)]. GI was ineffective for BDPN mapping [see Figs. 10(f) and 11(k)], as it showed lots of misclassification [see Figs. 10(k) and 11(f)]. DPNI achieved similar mapping results as RF in areas with complete DPN coverage [e.g., red rectangle in Fig. 10(h) and (l)]. DPNI had better results than RF in the areas where the two types of DPNs are mixed [e.g., red rectangle in Figs. 10(d) and (g) and 11(d) and (g)]. Moreover, RF had better performance than DPNI in the areas sparsely covered with DPNs [e.g., green rectangle in Fig. 10(d) and (g)]. Overall, DPNI can accurately identify DPNs and effectively suppress the backgrounds. It is satisfactory that DPNI achieved similar and better DPN mapping performance than RF in most areas.

V. Discussion

A. Stability of Threshold

Table IV shows the measurement results of DPNI threshold stability. Specifically, we evaluated DPN mapping accuracy using thresholds $-50\%\sim50\%$ off from the optimal threshold. DPNI demonstrated stability across all test cases. For Beijing, in spring, summer, and winter, the reduction of OA is less than 0.56%, and the reduction of $F1$-score is less than 1.86%, even with the threshold $\pm50\%$ off from the optimal threshold. For Zhengzhou, the reduction of OA is less than 0.66%, and the reduction of $F1$-score is less than 2.16%. In the autumn, with a threshold -50% off from the optimal threshold, $F1$-score reached 83.67% in Beijing and 92.52% in Zhengzhou. It can be illustrated that DPNI may be sensitive when the threshold is less than the optimal threshold in the fall, but the DPN mapping accuracy using DPNI is still satisfactory. In Beijing, DPNI achieved an accuracy close to the optimal accuracy with a threshold $\pm20\%$ off from the optimal threshold in spring and autumn and $\pm50\%$ off from the optimal threshold in summer and winter. In Zhengzhou, DPNI achieved an accuracy close to the optimal accuracy with a threshold $\pm20\%$ off from the optimal threshold in all four seasons.

We suggested using the threshold ranging from 80% to 120% of the optimal thresholds (i.e., 0.008–0.012 in spring and 0.024–0.036 in autumn) and 50%–150% (i.e., 0.005–0.015 in summer and winter) to map DPNs in areas where the main DPN type is GDPN (e.g., Beijing). The threshold ranging from 80% to 120% (i.e., 0.016–0.024 in spring and summer and 0.024–0.036 in autumn and winter) of the optimal thresholds are recommended to use to map DPNs in areas where the main DPN type is BDPN (e.g., Zhengzhou).

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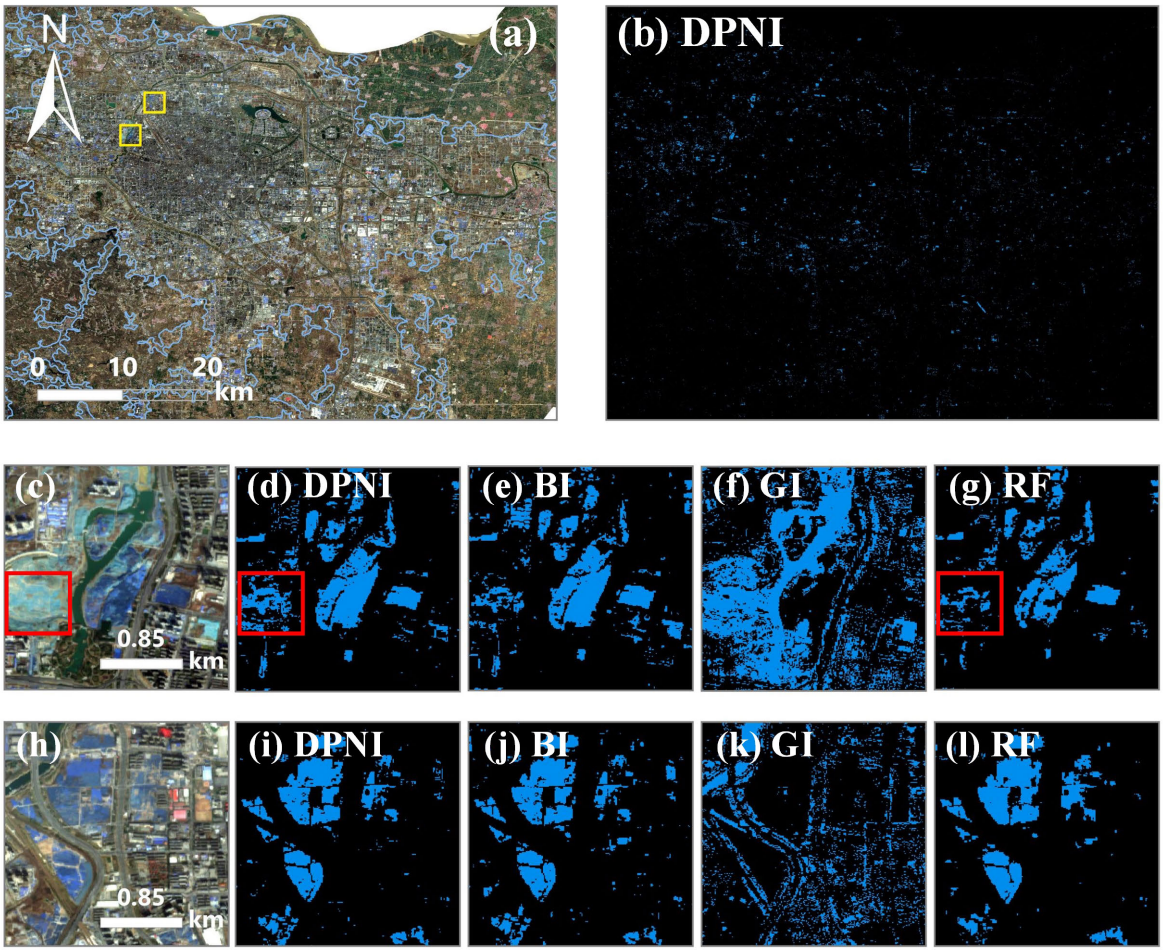


Fig. 11. DPN mapping results using different indices and the RF method in Zhengzhou. The yellow rectangle in (a) is the example region. (c) and (h) Sentinel-2 images of example regions. (b), (d), and (i) DPNI. (e) and (j) BI. (f) and (k) GI. (g) and (l) RF.

B. Stability of DPNI in Other Cities

The effectiveness of DPNI in mapping DPNs in other cities has been validated. The middle value of the recommended threshold range was applied to map DPNs in Tianjin, Zhoukou, Urumqi, and Hangzhou. The DPNs in Tianjin, Urumqi, and Hangzhou are mainly GDPNs, while those in Zhoukou are BDPNs. Fig. 12 shows the DPN mapping results in Tianjin, Zhoukou, Urumqi, and Hangzhou for spring, summer, autumn, and winter, utilizing the optimal thresholds identified in Beijing and Zhengzhou. Excitingly, DPNI is robust in different cities. More importantly, DPNI does not require samples and optimization of parameters, which is a significant advantage over classifier-based methods.

We obtained 3000 non-DPN samples in Tianjin, Zhoukou, Urumqi, and Hangzhou using the fishnet-based random sampling method mentioned in Section III-C. A total of 400 and 300 DPN samples were collected in Tianjin and Zhoukou, respectively, while a total of 100 and 200 DPN samples were collected in Urumqi and Hangzhou, respectively. These samples were used to quantitatively analyze the mapping accuracy of DPNI on DPNs. Table V illustrates the DPN mapping accuracy using DPNI with the middle value of the recommended threshold range. It can be found that DPNI achieves satisfactory mapping accuracy in all four seasons. The OA of the DPN mapping is 93.07%–99.30% for the GDPN and 96.24%–99.62% for the BDPN. Therefore, DPNI is still effective for different types of DPNs in other cities.

C. Effectiveness of DPNI to DPNs Covering Different Construction Materials

The construction materials covered by the DPNs can influence DPN spectral characteristics. Field spectral data of DPNs covering different construction materials demonstrated the effectiveness of DPNI. Fig. 13(a1) shows the results of the field spectral measurements as well as the resampled Sentinel-2 spectra. It can be found that the reflectance of different DPN types covering different materials is very similar in shape and magnitude. As shown in Fig. 13(a2), the materials covered by DPNs chiefly affect the magnitude, not the trend, of the spectral reflectance.

We computed the DPNI values for the resampled Sentinel-2 spectra of the DPNs. Fig. 13(a3) shows the DPNI values of DPNs covering different construction materials, as well as the mean DPNI values of other land cover samples (as mentioned in Section III-A). The results show that DPNs covering various materials demonstrated excellent separability from other land covers. Though water and DPNs have similar DPNI values, they are still distinguishable, as water has DPNI values less

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TABLE IV
DPN MAPPING ACCURACY USING DPNI WITH THRESHOLDS −50%∼50% OFF FROM THE OPTIMAL THRESHOLD

		-50 %	-40 %	-30 %	-20 %	-10 %	Optimal threshold	+10 %	+20 %	+30 %	+40 %	+50 %
	UA	86.23	87.41	88.91	90.01	90.68	92.76	94.26	94.55	95.24	95.54	**95.90**
Spring	PA	**89.15**	88.23	88.08	87.47	87.02	86.10	85.04	83.97	83.06	81.69	81.23
Beijing	OA	94.90	94.97	95.18	95.27	95.33	**95.46**	95.43	95.39	95.34	95.18	95.16
	F1-Score	82.54	82.57	83.14	83.31	83.43	**83.61**	83.56	83.00	82.69	81.91	81.75
	UA	94.45	94.87	95.01	95.29	95.29	95.86	95.86	96.14	96.12	96.13	**96.26**
Summer	PA	**95.17**	95.02	94.86	94.71	94.86	94.41	94.41	94.11	93.50	93.50	93.35
Beijing	OA	98.40	98.45	98.45	98.47	98.49	**98.52**	98.52	98.52	98.43	98.43	98.42
	F1-Score	94.81	94.94	94.93	95.00	95.08	**95.13**	95.13	95.11	94.79	94.79	94.78
	UA	80.84	85.37	87.92	89.84	91.26	93.00	93.29	94.14	94.03	94.64	**94.89**
Autumn	PA	**86.71**	86.40	85.80	85.50	85.20	84.29	83.99	82.48	80.97	80.06	78.55
Beijing	OA	97.19	97.64	97.84	97.99	98.09	**98.17**	98.17	98.12	97.99	97.97	97.87
	F1-Score	83.67	85.88	86.85	87.62	88.12	**88.43**	88.39	87.92	87.01	86.74	85.95
	UA	93.77	94.19	94.77	95.21	95.65	96.56	96.68	96.82	97.13	97.27	**97.59**
Winter	PA	**89.45**	89.15	88.85	88.85	88.26	87.52	86.48	85.88	85.59	84.84	84.25
Beijing	OA	97.44	97.46	97.51	97.53	97.55	**97.58**	97.44	97.37	97.37	97.27	97.23
	F1-Score	91.56	91.60	91.71	91.76	91.81	**91.82**	91.29	91.02	90.99	90.63	90.43
	UA	94.97	95.29	95.61	95.93	96.56	97.22	97.19	97.19	**97.50**	97.48	97.47
Spring	PA	**94.33**	94.33	94.33	94.33	93.67	93.33	92.33	92.33	91.00	90.33	90.00
Zhengzhou	OA	98.22	98.28	98.33	98.39	98.39	**98.44**	98.28	98.28	98.11	98.00	97.94
	F1-Score	94.65	94.81	94.97	95.13	95.09	**95.24**	94.70	94.70	94.14	93.77	93.59
	UA	92.31	92.60	92.88	93.18	93.79	94.39	94.31	94.28	**94.58**	94.52	94.46
Summer	PA	**95.68**	95.68	95.35	95.35	95.35	95.02	93.69	93.02	92.69	91.69	90.70
Zhengzhou	OA	97.94	98.00	98.00	98.06	98.17	**98.22**	98.00	97.89	97.89	97.72	97.56
	F1-Score	93.96	94.12	94.10	94.25	94.56	**94.70**	94.00	93.64	93.62	93.09	92.54
Autumn	UA	87.10	88.62	90.24	91.64	91.64	92.21	92.19	92.72	93.01	92.99	**93.29**
Zhengzhou	PA	**98.67**	98.34	98.34	98.34	98.34	98.34	98.00	97.34	97.34	97.01	97.01
	OA	97.33	97.61	97.94	98.22	98.22	**98.33**	98.28	98.28	98.33	98.28	98.33
	F1-Score	92.52	93.23	94.12	94.87	94.87	**95.18**	95.01	94.97	95.13	94.96	95.11
	UA	93.06	93.31	94.21	94.52	94.52	95.13	95.13	95.41	95.72	95.72	**96.03**
Winter	PA	**98.66**	97.99	97.99	97.99	97.99	97.99	97.99	97.32	97.32	97.32	96.99
Zhengzhou	OA	98.55	98.50	98.66	98.72	98.72	**98.83**	98.83	98.78	98.83	98.83	98.83
	F1-Score	95.78	95.60	96.06	96.22	96.22	**96.54**	96.54	96.36	96.52	96.52	96.50

TABLE V
DPN MAPPING ACCURACY USING DPNI WITH RECOMMENDED INITIAL THRESHOLDS IN TIANJIN, ZHOUKOU, URUMQI, AND HANGZHOU

City	Season	UA (%)	PA (%)	OA (%)	F1-Score (%)	Threshold
Tianjin	Spring	98.78	82.26	93.70	87.76	0.01
	Summer	98.71	97.22	98.93	97.96	0.01
	Autumn	94.54	78.16	94.01	89.91	0.03
	Winter	98.03	88.38	93.07	88.57	0.01
Zhoukou	Spring	95.19	100	99.62	97.54	0.02
	Summer	90.40	100	98.78	94.96	0.02
	Autumn	91.51	100	96.24	89.39	0.03
	Winter	93.52	95.59	98.90	94.55	0.03
Urumqi	Spring	88.23	88.23	98.60	88.23	0.01
	Summer	85.71	83.33	98.62	84.51	0.01
	Autumn	92.86	81.25	98.76	86.67	0.03
	Winter	83.33	93.75	98.74	88.23	0.01
Hangzhou	Spring	86.79	91.39	98.78	89.03	0.01
	Summer	84.61	99.00	99.30	91.24	0.01
	Autumn	85.48	100	98.34	92.17	0.03
	Winter	87.00	96.67	98.41	91.58	0.01

than 0, whereas DPNs have the opposite [see Fig. 13(a4)]. Therefore, DPNI is effective for the variation in construction materials covered by DPNs under the spectral resolution of Sentinel-2 data.

D. Advantages and Limitations of DPNI

In this study, a new spectral index named DPNI was developed using Sentinel-2 data that can highlight the spectral differences between DPNs and other land covers. The

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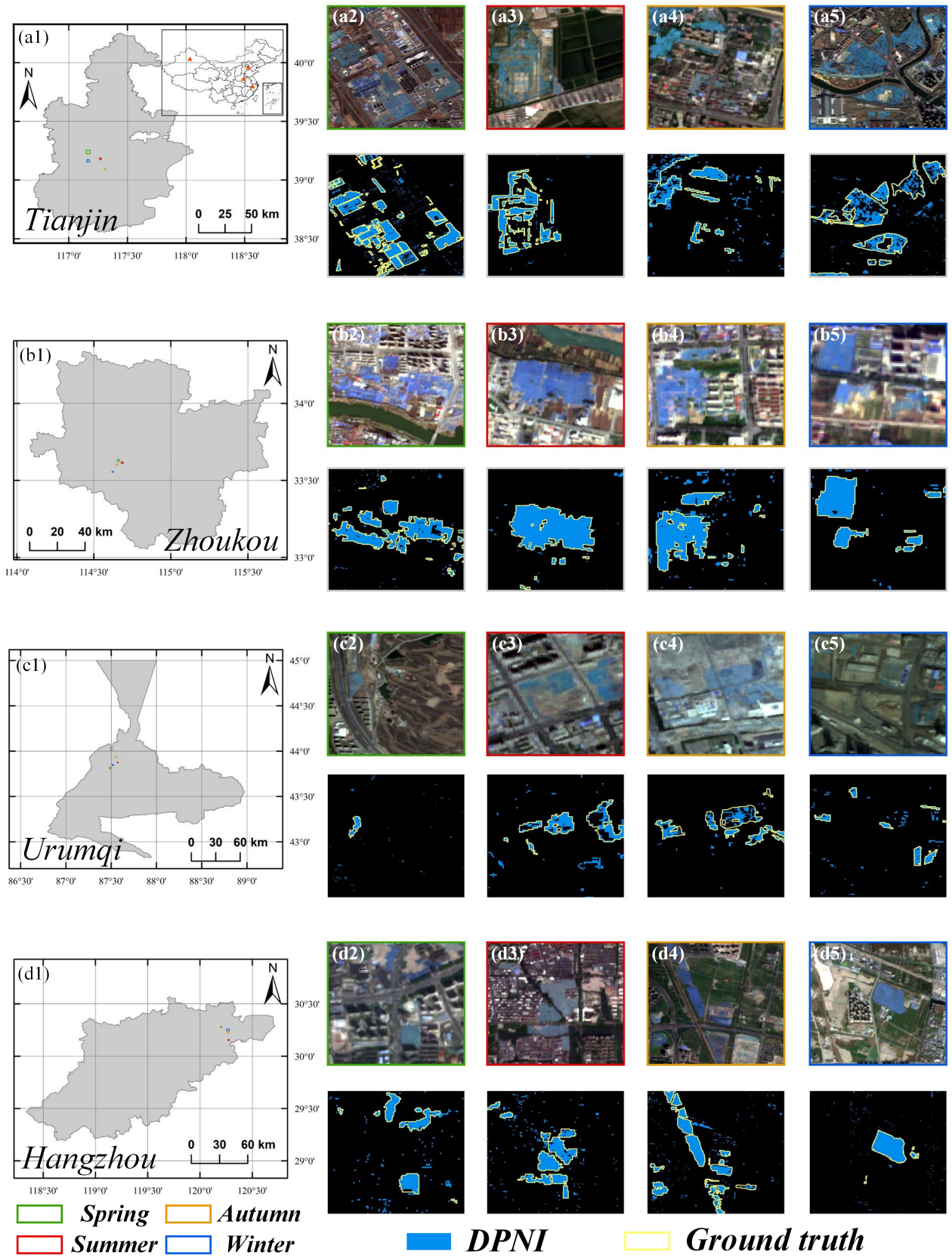


Fig. 12. DPN mapping results for different cities using recommended initial thresholds generated in Beijing and Zhengzhou. For (a1)–(a5) Tianjin, (b1)–(b5) Zhoukou, (c1)–(c5) Urumqi, and (d1)–(d5) Hangzhou in spring, summer, autumn, and winter, respectively.

experiment results showed that the performance of DPNI is better than the compared indices and is close to the classifier-based method. The strengths of DPNI can be concluded as follows.

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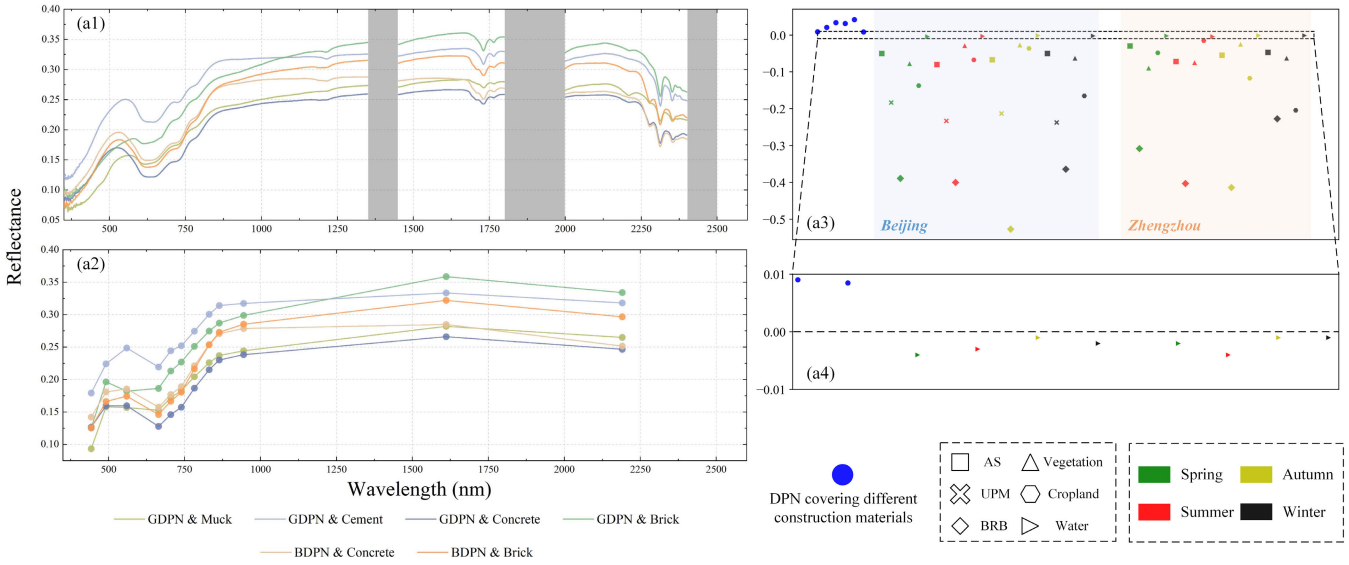


Fig. 13. (a1) Field reflectance spectra of six types of DPNs. (a2) Resampled Sentinel-2 spectra of six types of DPNs. (a3) and (a4) Distribution of DPNI values for DPNs covering different construction materials, as well as the mean DPNI values for other land cover samples in the study areas. The gray regions of 1350–1450 nm, 1800–1999 nm, and 2401–2500 nm in (a1) denote the noise regions.

First, although the urban environment is complex, DPNI can suppress different backgrounds and effectively highlight DPNs. The established goal of the DPNI is to separate DPN spectra from non-DPN spectra. The $B2$–$B4$ indicator and the $B7$–$B2$ indicator are used in DPNI to suppress UPMs and BRBs, respectively. The $B2$ and $B12$ are used to suppress vegetation, cropland, and water. The spectral separability demonstrates that these methods used to suppress backgrounds are effective. DPNI is the first spectral index to identify DPNs using spectral separability.

Second, DPNI is effective for all types of DPNs in different cities. Although there are spectral differences between GDPNs and BDPNs, the proposed DPNI can effectively identify both GDPNs and BDPNs. More importantly, the field spectral analysis illustrates that DPNI is effective for the variation in construction materials covered by DPNs.

Finally, DPNI provides a simple and effective method for DPN mapping. Classifier-based methods for DPN mapping usually have two main problems: 1) classifier-based methods require a sample collection containing all classes. It is challenging to collect comprehensive samples of non-DPN land covers in urban areas. 2) Classifier-based methods usually require optimizing parameter settings continuously, which is complex and calculator-intensive, limiting the large-scale DPN mapping. DPNI, as an index-based method, requires no input parameters. Crucially, although no samples are used to optimize the threshold, the accuracy of the DPN mapping results using DPNI in different cities is still satisfactory, as the threshold of DPNI shows stability. Regardless of whether the recommended threshold from the study area is used to map DPN in other cities, or $-50\%\sim50\%$ off from the optimal threshold is used within a single area, DPNI can still achieve substantial mapping accuracy. In addition, DPNI has proven to be robust and effective for DPN mapping using Sentinel-2 data. Sentinel-2 is the most popular accessible satellite data and provides global coverage of land surfaces. Therefore, DPNI has good application potential for DPN mapping on a national and global scale.

Although DPNI has produced satisfactory results, there are still some limitations. First, the effectiveness of DPNI applied to other satellite data requires additional validation. Second, it is difficult to consider the influence of complex environmental factors on the spectral reflectance of DPNs. For example, the DPN spectra can be influenced by the different construction materials covered with the DPNs. Finally, although DPNI demonstrates robustness to various construction materials under the spectral resolution of Sentinel-2, the influence of different construction materials on the spectral characteristics of DPNs still merits further investigation. Therefore, very high-resolution hyperspectral data will be used for exploration in the future (e.g., WorldView-3).

VI. Conclusion

Previous studies have proposed some DPN mapping methods, but they are all classifier-based methods, which are black boxes. No spectral index has been developed for DPN mapping, and the investigation of the spectral characteristics of DPN is still lacking. To fill this gap, a new robust spectral index, named DPNI, was proposed for rapid and accurate mapping of DPN using Sentinel-2 data in this study. According to the separability analysis, the $B2$–$B4$ indicator and the $B7$–$B2$ indicator were used to separate DPNs from AS, UPMs, and BRBs. The blue and SWIR2 bands were chosen as indicators to distinguish DPNs from land covers, including cropland, vegetation, and water. The DPNI was evaluated in urban areas with complex environments and was compared with established index-based (BI and GI) and classifier-based (RF) methods. The visual evaluation and accuracy assessment results showed that the DPNI distinguishes between DPNs and other land covers more effectively than BI and GI. DPNI achieved an OA of 93.51%–98.83% and an $F1$-score of 83.61%–96.54% in different seasons, which showed significant

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advantages for DPN mapping. The mapping results based on DPNI and RF had a similar level. However, the advantage of the DPNI was its independence from samples and parameter settings, enabling rapid, large-scale DPN mapping. DPNI thresholds used for DPN mapping had good stability for various kinds of DPNs in different seasons. Furthermore, DPNI was robust in different cities, and satisfactory DPN mapping results can be obtained directly using our recommended thresholds. The field spectral analysis demonstrated the effectiveness of DPNI for DPNs covering different construction materials. Overall, DPNI is the first spectral index of the DPN, which provides a simple, efficient, and generalizable method for DPN mapping. DPNI is expected to advance automatic and large-scale DPN mapping methods and support the assessment of the environmental impacts of DPNs.

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